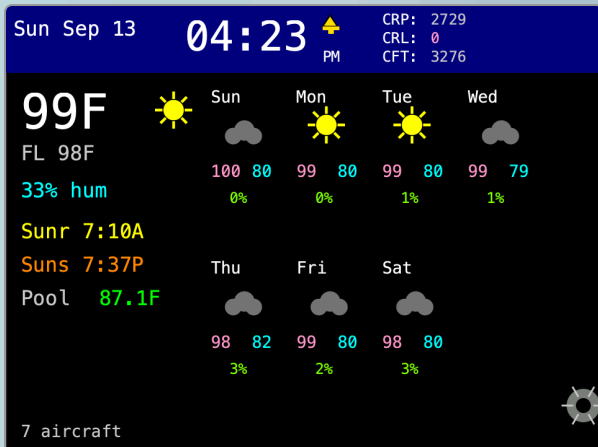


What It Shows

- **Alarms** — up to 6 weekday-scheduled alarms with an LED + sound alert.
- **Clock & date** — always on top.
- **Firmware updates** — checks for new firmware once a day and installs it over WiFi, or update manually.
- **Flight recall** — tap the aircraft count to bring the last overhead flight's details back up, even after it's gone.
- **Flights overhead** (optional — on by default) — a mini radar of live aircraft near you: callsign, route, altitude, speed, distance, airline logo, and the plane's actual ground track.
- **Pool temperature** (optional — off by default) — current water temperature and history graphs, if you have a Govee pool thermometer.
- **Settings** — every option on the device is configurable on-screen: units, clock, WiFi, location, flight tracking, alarms, and more.
- **Sleep mode** — the screen sleeps overnight and wakes on touch.
- **Startup chime** — a short rising melody + LED sweep plays on power-on (stays silent when it wakes from sleep mode). Additionally, a chime plays after a firmware update.
- **Weather** — current temperature, feels-like, humidity, sunrise/sunset, a 7-day forecast, and temperature history graphs (Day / Week / Month / Year).



Resetting (Settings → Reset)

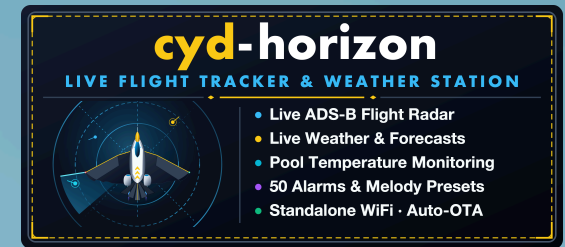
Every option asks you to confirm and says exactly what it clears:

- **Restart** — reboots only; clears nothing.
- **Graph Data** — clears pool and weather temperature history.
- **Settings** — clears settings and credentials (keeps touch calibration).
- **Factory Reset** — clears *everything*: settings, credentials, files, history, and touch calibration. The next boot recalibrates touch and asks for WiFi again.
- **Cancel** — backs out without changing anything.

⚠ **Factory Reset erases any pre-loaded airline logos** — restoring logos requires regenerating and reloading them from a development computer.

Troubleshooting

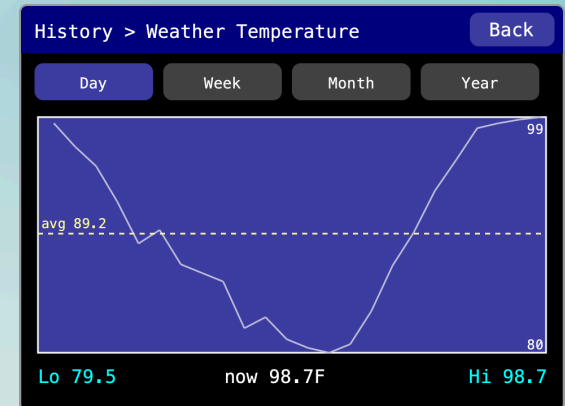
- **Taps land in the wrong spot** — press and hold anywhere on the screen for **10 seconds** to recalibrate.
- **Wrong weather or no flights nearby** — fix your location under **Settings → Location**.
- **A flight's route or track looks wrong** — route and ground-track data come from public flight-data feeds that may not have the latest data, so they can be missing, outdated, or inaccurate.
- **New WiFi network** — re-enter it under **Settings → Network**.



A small self-contained touchscreen dashboard for **Cheap Yellow Display (CYD)** boards:

- **2.8" 2432S028R**
- **4" E32R40T**

Once it's set up it runs on your WiFi — no computer needed.



Firmware v3.5.1+

github.com/improving-minnesota/cyd-horizon

Getting Started

1. **Plug it in.** On a brand-new device (or after a Factory Reset) it first asks you to tap a few crosshairs to calibrate the touchscreen — this is skipped if calibration is already saved.
2. **Enter WiFi.** Type your network name and password with the on-screen keyboard. (Change it later under **Settings → Network**.)
3. Done — the dashboard appears and starts loading weather and flights.

Set Your Location

Weather and flights need to know where you are. On first boot the device guesses your location from your internet connection — check that it's right:

Open **Settings → Location**, then use whichever is easiest:

- **Search Address** — type your city or address and pick the match.
- **Set** — type your latitude and longitude directly.
- **Find by IP** — guess again from your connection.

OpenSky Account (Recommended)

Flights work out of the box, but a free OpenSky account raises your daily limit and removes the yellow "anonymous" warning.

1. Create a free account at **opensky-network.org**.
2. Go to **My OpenSky → Account** and create an **API client** — that downloads a file containing your **client ID** and **client secret**.
3. On the device: open **Settings → Flight Tracker**, page right to the **OpenSky Credentials** screen, and enter both.

The three small readouts in the header — **CRP**, **CRL**, **CFT** — show your remaining daily OpenSky credits (tap them for detail). Grey is healthy, yellow is low, pink is nearly out. Credits reset daily.

Govee Pool Thermometer (Optional)

Only if you own a Govee pool thermometer:

1. Create a free developer account at **developer.govee.com** and generate an **API key**.
2. On the device: open **Settings → Pool Temp**, turn it **on**, enter the key, tap **Fetch Devices**, and pick your thermometer.

Using the Dashboard

The main screen isn't just a display — several spots are **buttons you can tap**:

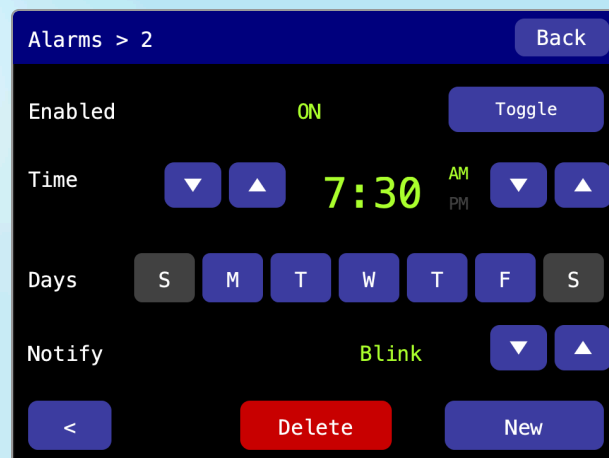
- **Cog (bottom-right)** — open settings.
- **Weather temperature (top-left)** — weather history graph.
- **Pool reading** — pool temperature history graph.
- **Aircraft count (bottom-left)** — details of the last overhead flight.
- **CRP / CRL / CFT (header)** — OpenSky credits detail.
- **Clock/date (header)** — open the alarm editor.
- Any settings or graph screen returns to the dashboard after 2 minutes untouched.

Alarms

Up to 6, each with its own weekdays and a **Notify** pattern. Leave all weekdays off for a one-time alarm. Set the time with the ▼ / ▲ arrow buttons — hour on the left, minute on the right (the hour rolls through AM/PM) — or tap the time to type it. A bell icon sits beside the clock while any alarm is on. **Delete** asks for confirmation before removing an alarm.

When one fires: **Dismiss** stops it for today; **Snooze** refires in 5 min (up to 3 snoozes, then it dismisses for the day). Alarms fire on time even during Sleep Mode — the device wakes itself at the alarm time — and a snoozed or missed alarm still fires after a power loss.

 Sound requires an **external speaker** plugged into the board's small JST port — the display has no built-in speaker, so without one alarms flash the LED only. Loudness comes from **Notify Volume** under Settings → General; it applies to alarms, the callsign alert, and the boot chime.



Alarms > 2

Enabled ON Toggle

Time ▼ ▲ 7:30 AM ▼ ▲ PM

Days S M T W T F S

Notify Blink ▼ ▲

< Delete New

Settings at a Glance

Defaults shown in parentheses.

- **About** — version, author, and firmware update status / install.
- **Calibrate Touch** — rerun touch calibration.
- **Flight Tracker** — enabled (on); radar radius (3.5, in mi/km/nm — how far away to look); altitude ceiling (15,000, in ft/m — planes above it are ignored); poll interval (30 s); timer bar (off); home airport (none); watch callsign — (none, matches any part of a callsign; * matches every flight); callsign notify (Radar); LED blink (on); Show IATA Airports (on); OpenSky credentials.
- **General** — theme color picked from swatch rows (tap a hue, then a shade or grey — colors the header and screen buttons); auto-update (on); Notify Volume (levels 1–10, default 10); units (Imperial ft/mpH/mi, Metric m/kts/km, or Aviation ft/kts/nm — also sets weather and pool temps to °F or °C); clock (12-hour with AM/PM, or 24-hour).
- **Location** — your coordinates (auto-guessed on first boot).
- **Network** — WiFi network (shows *Scanning* while it searches); DHCP or static IP; device hostname.
- **Pool Temp** — (off) enable, enter the API key, pick your thermometer. While off, no data is collected.
- **Reset** — see the back cover.
- **Sleep Mode** — enabled (on); sleeps 10:00 PM – 8:00 AM; a touch wakes it for 5 minutes.

LED & Border Colors

- **Red border / LED** — something needs attention: no WiFi, bad OpenSky credentials, exhausted OpenSky credits, or missing weather/pool data.
- **Yellow** — OpenSky is running anonymously; flights still work.
- **Flight blinks** — blue for any overhead flight; red when it's departing your home airport; green when arriving; yellow when both ends are your home airport. A watched callsign instead plays its **Callsign Notify** pattern on the LED and speaker while its details are shown — independent of the blink setting.

Firmware Updates

The device keeps its firmware current over WiFi. **Auto-Update** (on by default) checks once a day and installs new releases; to update manually, open **Settings → About** and tap **Install** if a newer version is offered. If an update fails, it rolls back automatically.